

🚀 Deployment Guide — React + Node.js + MongoDB App

A simple step-by-step guide to deploy any React frontend + Node.js backend project.

📦 What You Need

- GitHub account
- Vercel account (vercel.com)
- Render account (render.com)
- MongoDB Atlas account (mongodb.com)

📋 Overview

...

Frontend → Vercel (free)
Backend → Render (free)
Database → MongoDB Atlas (free)

...

STEP 1 — Push Project to GitHub

```
``bash
git init
git add .
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/yourusername/your-repo.git
git push -u origin main
``
```

> If repo already exists and push is rejected:

```
> ``bash
> git pull origin main --rebase
> git push origin main
> ``
```

STEP 2 — Deploy Backend on Render

1. Go to `https://render.com` → sign in with GitHub
2. Click **"New +"** → **"Web Service"**
3. Connect your GitHub repo
4. Fill in settings:

...

Name → your-app-backend
Region → Singapore (for India)
Branch → main
Root Directory → backend

Build Command → npm install

Start Command → npm start

Instance Type → Free

...

5. Scroll down → **Environment Variables** → add:

...

MONGO_URI → your mongodb connection string

JWT_SECRET → any random secret string

PORT → 5000

...

6. Click **Deploy Web Service**

7. Wait 3-5 minutes

8. Copy your Render URL:

...

<https://your-app-backend.onrender.com>

...

STEP 3 — Update Frontend API URL

Open `frontend/src/App.jsx` (or wherever your API URL is):

```
``javascript
```

```
// Change this
```

```
const API_BASE = 'http://localhost:5000/api';
```

```
// To this
```

```
const API_BASE = 'https://your-app-backend.onrender.com/api';
```

```
...
```

Then push to GitHub:

```
``bash
```

```
git add .
```

```
git commit -m "update API URL to production"
```

```
git push origin main
```

```
...
```

STEP 4 — Deploy Frontend on Vercel

1. Go to `https://vercel.com` → sign in with GitHub

2. Click **Add New Project**

3. Import your GitHub repo

4. Fill in settings:

...

Framework Preset → Vite (or React)

Root Directory → frontend

...

5. Click **Deploy**

6. Wait 2 minutes

7. Your live URL:

```
```\nhttps://your-app.vercel.app\n```
```

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## ## STEP 5 — Update CORS in Backend

Open `backend/server.js`:

```
```\njavascript\n// Change this\napp.use(cors({ origin: '*' })))\n\n// To this\napp.use(cors({\n  origin: 'https://your-app.vercel.app',\n  methods: ['GET', 'POST', 'PUT', 'DELETE'],\n  allowedHeaders: ['Content-Type', 'Authorization']\n})))\n```
```

Push the change → Render auto redeployes.

STEP 6 — MongoDB Atlas Setup

1. Go to `https://mongodb.com/atlas`
2. Create a free cluster
3. Click **"Connect"** → **"Drivers"** → copy connection string
4. Go to **"Network Access"** → Add IP → `0.0.0.0/0` (allow all)
5. Paste connection string in Render environment variables as `MONGO\_URI`

🔄 How to Update Your App Later

Every time you make changes:

```
```\nbash\ngit add .\ngit commit -m "describe your change"\ngit push origin main\n```
```

- **\*\*Vercel\*\*** auto redeployes frontend ✅
- **\*\*Render\*\*** auto redeployes backend ✅

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## ## ⚠️ Common Problems & Fixes

Problem	Fix
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Vercel shows 404	Set Root Directory to `frontend` in Vercel Settings
Push rejected	Run `git pull origin main --rebase` then push again
Backend not connecting	Check MONGO\_URI in Render environment variables
CORS error in browser	Update `origin` in server.js with your Vercel URL
Render sleeping	First request takes ~30 sec on free plan, normal behavior
API not working	Make sure API\_BASE in App.jsx points to Render URL not localhost

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## ## ✅ Final Checklist

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- ☐ Code pushed to GitHub
- ☐ Backend deployed on Render
- ☐ Environment variables added in Render
- ☐ MongoDB Atlas IP whitelist set to 0.0.0.0/0
- ☐ API\_BASE updated in frontend with Render URL
- ☐ Frontend deployed on Vercel
- ☐ Root Directory set to frontend in Vercel
- ☐ CORS updated with Vercel URL in backend
- ☐ .env added to .gitignore

...

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## ## 🔒 Important — Never Push These to GitHub

Add these to `.gitignore`:

...

.env  
node\_modules/  
dist/  
...

Your `.gitignore` should have:

...

# Environment variables  
.env  
.env.local

# Dependencies  
node\_modules/

# Build output  
dist/  
build/  
...

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\*Keep this file handy for your next project deployment!\* 🚀